CAPSTONE PROJECT · DEPI DATA ANALYSIS TRACK

THE EGYPTIAN EXPORT OBSERVATORY

Egypt Trade Intelligence Dashboard — 2010 to 2024

Project Guidebook

Powered by Microsoft Power BI · 8 Dashboard Pages · 40+ DAX Measures

Data Analysis DEPI Track	**2010 – 2024** Data Period	**8 Pages (P1–P8)** Dashboard Pages	**CAPMAS · Comtrade · WTO** Data Sources

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01 Executive Summary

★ **Egypt exported $32.6 billion in 2024 against a Vision 2030 target of $104 billion — a 31% achievement rate with only 6 years remaining.**
★ **The trade deficit stands at -$50 billion and has widened every year since 2010.**
★ **Over $3 billion in annual export revenue is lost to systematic underpricing — without needing to grow volumes.**
★ **62% of exports remain petroleum-dependent, exposing the economy to commodity price cycles.**
★ **Vietnam — which had a lower GDP per capita than Egypt in 2000 — today exports 11× more.**

The Egyptian Export Observatory is an 8-page interactive intelligence dashboard built in Microsoft Power BI, covering Egypt's foreign trade performance across a 15-year horizon (2010–2024). Drawing on official data from CAPMAS, UN Comtrade, the WTO, the World Bank, and ITC TradeMap, the Observatory transforms over 7,000 records across 17 structured data tables into a decision-ready analytical platform.

The project identifies four structural gaps: a chronic trade deficit; a petroleum-dominated export basket; systematic export underpricing; and geographic concentration risk. The Observatory provides not only diagnosis but a precise roadmap for action — 12 data-driven recommendations, each quantified with projected impacts, timelines, and evidence bases drawn directly from the underlying data.

$32.6B	-$50B	31%	$3B+	120+	55%
Total Exports 2024	Trade Deficit 2024	Vision 2030 Achievement	Annual Revenue Lost	Active Trade Partners	Non-Oil Export Share

02 Project Overview & Context

What is the Egyptian Export Observatory?

The Egyptian Export Observatory is a data analytics capstone project developed under the Digital Egypt Pioneers Initiative (DEPI) by a team of five analysts. It is an 8-page interactive Power BI dashboard that consolidates 15 years of Egypt's official trade data (2010–2024) into a unified intelligence layer — making complex trade statistics accessible, visual, and actionable.

Think of it as Egypt's "trade health monitor." Just as a hospital monitor shows vital signs at a glance, the Observatory shows the vital signs of Egypt's export economy: how much we export, to whom, at what price, compared to what target, and what it would take to reach Vision 2030.

Project Attribute	Detail
Project Name	The Egyptian Export Observatory — Egypt Trade Intelligence Dashboard
Program	Digital Egypt Pioneers Initiative (DEPI) — Data Analysis Track
Dashboard Tool	Microsoft Power BI (Desktop + Service)
Data Coverage	2010–2024 (15 Years of Official Trade Records)
Total Records	~7,000+ rows across 17 structured tables
Dashboard Pages	8 Pages (P1 Executive Summary through P8 Scenario Analysis)
DAX Measures	40+ professional measures across 7 analytical categories
Data Sources	CAPMAS 2024 · UN Comtrade · WTO · World Bank · ITC TradeMap

Why Does This Project Exist?

Egypt possesses structural trade advantages that most countries would envy: a population of 105 million, the Suez Canal — one of the world's most strategic waterways — proximity to the EU and MENA markets, abundant agricultural land, and a young labor force. Yet Egypt's export performance chronically underperforms its potential.

The core problem is not a lack of data — Egypt has rich official statistics from CAPMAS and strong international trade databases. The problem is that this data is fragmented, inaccessible to most users, and never visualized in a way that makes the opportunities and gaps immediately clear. The Observatory was built specifically to close this information gap.

03 Problem Statement — Four Structural Gaps

Core Question: Why does Egypt consistently underperform its export potential — and where exactly are the losses occurring?

Gap 1: The Widening Trade Deficit

Egypt's trade deficit expanded from -$29 billion in 2010 to -$50 billion in 2024. While exports grew at a 2.7% CAGR, imports grew at 3.9% — meaning the structural gap is not narrowing; it is accelerating. To understand why this matters: a persistent trade deficit means Egypt is continuously spending more foreign currency than it earns from exports. Over time, this creates pressure on the Egyptian pound, drains foreign exchange reserves, and limits the government's capacity to invest in public services.

The 2022 export surge of 34% appears impressive but was driven entirely by global commodity price spikes (primarily LNG and petroleum), not by new products or markets. Once prices normalized, exports fell back — confirming the structural weakness was never fixed.

Gap 2: Petroleum Dependency & Low Export Complexity

62% of Egypt's exports are petroleum and natural gas products. The remaining 38% — primarily agricultural commodities, textiles, fertilizers, and basic chemicals — score an Export Sophistication Index (EXPY) of approximately 10,800. To put this in context: Turkey, with comparable geographic advantages, scores 19,500; Vietnam scores 15,200.

EXPY measures how "sophisticated" a country's exports are — essentially, what kind of products rich countries export. Egypt is primarily exporting raw and semi-processed materials. Countries that export manufactured, branded, high-value-added products earn far more per unit — and are far less vulnerable to commodity price cycles. Vision 2030 sets a target of 75% non-oil export share; Egypt is currently at 55%.

Gap 3: Systematic Export Underpricing — $3 Billion Lost Annually

This is perhaps the most actionable finding in the entire Observatory. Egypt sells its export commodities at prices 30–70% below market rates achieved by competitor nations. A concrete example: Egypt exports citrus at $333 per ton; Spain sells the same variety at $426 per ton; German importers pay $576 per ton. The margin between what Egypt receives and what the end market pays is captured entirely by intermediaries and foreign processors.

This is not a marginal inefficiency. Across all products, this pricing gap translates to over $3 billion in annual foregone revenue — revenue Egypt could capture without growing export volumes by a single ton. This is why the Price Intelligence module (Page 5) is arguably the most impactful page in the entire dashboard.

Gap 4: Geographic Concentration Risk

Egypt's top five trading partners account for 52% of total exports. Saudi Arabia and the UAE alone represent 22% of export value. Asia — which drives 60% of global GDP growth — absorbs only 18% of Egyptian exports. This geographic concentration means a diplomatic dispute, regional recession, or policy shift in just a few countries can significantly damage Egypt's entire export performance. Diversification is not just an opportunity — it is a risk management imperative.

Gap 5: Lagging Vision 2030 Progress

Egypt's Vision 2030 established 18 measurable trade and investment KPIs with specific 2030 targets. The average achievement rate across all KPIs as of 2024 is 43%. Six indicators are classified as critical. Achieving the $104 billion export target from the current $32.6 billion base requires an 18% compound annual growth rate — a rate that has never been sustained without deliberate structural reform. The Vision 2030 Tracker (Page 7) monitors all 18 KPIs with real-time progress bars and trend lines.

04 Project Objectives

Primary Objective

To design, build, and deliver a fully functional, data-driven export intelligence platform that enables evidence-based analysis of Egypt's trade performance — supporting strategic decision-making by policymakers, researchers, and the private sector in pursuit of Vision 2030 export targets.

Analytical Objectives

→ **Diagnose the Trade Deficit:** Quantify and contextualize Egypt's trade balance trajectory 2010–2024, separating structural drivers from cyclical (price) effects.

→ **Map Export Composition:** Decompose the export basket by sector and petroleum vs. non-petroleum classification to expose diversification gaps.

→ **Quantify the Pricing Gap:** Calculate the annual revenue loss from export underpricing across all product categories, benchmarked against competitor and importer-paid prices.

→ **Assess Geographic Diversification:** Evaluate partner country concentration using the Herfindahl-Hirschman Index (HHI) and identify underserved high-growth markets.

→ **Benchmark Peer Performance:** Compare Egypt's trade KPIs against Turkey, Morocco, and Vietnam across 14 indicators.

→ **Track Vision 2030 Progress:** Monitor achievement status across 18 official Vision 2030 KPIs, flagging critical gaps and computing the required CAGR to close them.

→ **Model Scenarios:** Build an interactive what-if scenario engine that quantifies potential revenue gains from addressing pricing, diversification, and import substitution gaps.

Technical Objectives

- Design a 17-table Fact Constellation data model with 7 fact tables and 6 dimension tables, optimized for Power BI performance.
- Author 40+ DAX measures across 7 analytical categories including growth metrics, price intelligence, benchmarking, and Vision 2030 tracking.
- Implement 7 custom tooltip pages providing contextual analytical depth across all 8 dashboard pages.
- Configure interactive features: drill-through, bookmarks, dynamic What-If parameters, and cross-filter interactions.
- Apply Egypt's national brand color palette — Navy / Emerald / Gold — consistently across all 8 pages.

05 Scope Definition

Understanding what is and is not included in the Observatory is critical for setting the right expectations. The scope was carefully defined to maximize analytical depth within the boundaries of available official data and the DEPI program timeline.

✓ IN SCOPE	✗ OUT OF SCOPE
✓ Egypt export & import flows, 2010–2024	✗ Real-time or live data feed integration
✓ 17 structured tables across 7 analytical domains	✗ Firm-level or sub-national trade data
✓ Product-level HS code classifications	✗ Customs clearance or logistics operational data
✓ Bilateral partner flows (120+ countries)	✗ Macroeconomic modelling beyond trade metrics
✓ Price gap analysis vs. competitor benchmarks	✗ Financial instrument or currency hedging analysis
✓ Vision 2030 KPI tracking (18 indicators)	✗ WTO legal or trade agreement compliance advisory
✓ Peer benchmarking: Turkey, Morocco, Vietnam	✗ Sector-level production cost modelling
✓ What-If scenario analysis and revenue projection	✗ Social or environmental impact assessment
✓ 40+ DAX measures and 8 Power BI pages	✗ Real-time news or sentiment monitoring

06 Stakeholders

The Observatory was designed with multiple user personas in mind — from a minister scanning KPI cards during a briefing, to an analyst drilling into bilateral trade flows, to a student exploring Egypt's trade data for the first time.

Stakeholder	Role	Engagement & Interest
Ministry of Trade & Industry	Primary Beneficiary	Dashboard insights inform export promotion policy and trade agreement negotiations.
CAPMAS (Central Agency for Statistics)	Data Partner	Provides official data; benefits from enhanced visualization of trade statistics.
Export Promotion Council	Operational Beneficiary	Uses product-level pricing and geographic analysis to target promotion efforts.
Egyptian Exporters (SMEs & Corporates)	End Beneficiary	Gain access to benchmark pricing, partner intelligence, and performance data.
DEPI Program (MCIT)	Program Sponsor	Evaluates educational outcomes and technical delivery quality of the capstone.
Academic & Research Community	Secondary Beneficiary	Methodology, data model, and analytical frameworks are replicable and publishable.
International Partners (WTO, World Bank)	Data Contributor	Contribute benchmark data; may use Egypt-specific insights for regional reporting.
Private Sector Investors & FDI	Indirect Beneficiary	Access to structured trade data reduces information asymmetry for investment decisions.

07 Dashboard Pages & Functional Requirements

The Observatory is structured as a logical story that flows from high-level overview to deep analytical detail. Each page answers a specific strategic question and hands off to the next with a clear narrative thread. Think of the 8 pages as 8 chapters of Egypt's trade story.

P1 Executive Summary

Key Question: *What is Egypt's trade situation right now?*

The landing page — designed for a minister who has 30 seconds. Four KPI cards (Total Exports, Trade Balance, Non-Oil Share %, Vision Achievement %) provide immediate situational awareness. A Trade Balance Waterfall Chart (2010–2024) shows the deficit trajectory in one visual. An Export vs. Import Area Chart reveals the structural divergence. A Vision 2030 KPI Progress indicator closes the page.

Key Measures: Total Exports USD M · Trade Balance USD M · Non-Oil Export Share % · Vision Achievement %

P2 Trade Trends 2010–2024

Key Question: *How have exports grown — and is it structural or cyclical?*

Builds the 15-year contextual story. A Multi-Line Chart separates oil from non-oil export trajectories, exposing the 2022 commodity price spike for what it was: a temporary windfall, not structural growth. A Ribbon Chart reveals how sector rankings have shifted over 15 years. A YoY Growth Bar Chart color-codes years where growth was commodity-driven vs. genuinely diversified.

Key Measures: YoY Export Growth % · CAGR Exports 2010–2024 · Non-Oil Export Share % · Cumulative Growth Index

P3 Geographic Intelligence

Key Question: *Where does Egypt export — and where should it export more?*

Maps Egypt's trade relationships globally. An Azure Maps filled map shows partner country distribution by export value intensity. A Treemap ranks the top 15 bilateral partners. A Sankey Diagram visualizes regional export and import flows simultaneously, making geographic concentration visible at a glance.

Key Measures: Count Active Partners · Geographic HHI · Exports to GAFTA · Exports to EU

P4 Product & Sector Deep Dive

Key Question: *What is Egypt actually exporting — and at what level of sophistication?*

Decomposes the export basket. A Donut Chart shows sector composition (petroleum vs. non-petroleum). A Horizontal Bar Chart displays Revealed Comparative Advantage (RCA) by product — products with RCA > 1 are where Egypt has a genuine competitive edge. A Scatter Plot maps product complexity against export value to identify the opportunity space for upgrading.

Key Measures: Top 5 Products Share % · Non-Oil Export Share % · RCA Products Count · Export HHI

P5 Price Intelligence

Key Question: *How much revenue is Egypt losing by underpricing its exports?*

The most impactful analytical module. A Revenue Loss Bar Chart ranks products by annual foregone revenue — the lost money is ranked from largest to smallest, making the priorities immediately obvious. A Dumbbell Chart compares Egypt's price to competitor price and final importer-paid price side by side. The circular trade analysis identifies products Egypt exports cheaply and then re-imports at a premium.

Key Measures: Total Revenue Lost USD M · Avg Price Gap % · Import Premium % · Circular Trade Count

P6 Peer Benchmarking

Key Question: *How does Egypt compare to Turkey, Morocco, and Vietnam?*

Quantifies the competitive gap. A Clustered Bar Chart enables KPI-by-KPI comparison across 14 indicators. A Radar Chart provides an overall performance profile — Egypt's shape vs. peers makes weaknesses instantly visible. A Journey Timeline charts each peer country's policy reform milestones that drove their export growth — providing Egypt with an evidence-based reform roadmap.

Key Measures: Egypt vs. Vietnam Gap % · Egypt EXPY Score · LPI Egypt · Export per Capita USD

P7 Vision 2030 KPI Tracker

Key Question: *Is Egypt on track to achieve its 2030 trade targets?*

Monitors Egypt's progress against 18 official Vision 2030 targets. Color-coded progress bars show achievement status: On Track (green), Progressing (amber), or Critical (red). A KPI Trend Line tracks any selected indicator dynamically. A countdown clock shows years remaining to the 2030 deadline. The Years to 2030 metric creates urgency and grounds all other analyses in the strategic timeline.

Key Measures: Vision KPI Achievement % · KPIs Critical Count · Gap to Vision Target · Years to 2030

08 Chapter 1: Data Preparation

Overview

Data preparation is the invisible foundation of every reliable dashboard. No matter how beautiful or sophisticated the visualizations are, if the underlying data is dirty, inconsistent, or poorly structured, the insights will be wrong — and wrong insights lead to bad decisions. In this project, data preparation was performed using Power Query (M Language) inside Microsoft Power BI Desktop.

Power Query is a data transformation engine that allows analysts to connect to data sources, clean and reshape the data, and load it into a structured model — all through a visual interface backed by a programming language called M. Every transformation is recorded as a step, making the process fully auditable and reproducible.

What Data Preparation Involved in This Project

- Connecting to 5 separate official data sources (CAPMAS, UN Comtrade, WTO, World Bank, ITC TradeMap).
- Loading and cleaning 17 tables totaling over 7,000 rows of trade data.
- Standardizing date formats, country name spellings, and currency units across all sources.
- Creating calculated columns (e.g., revenue loss per product, price gap per unit).
- Building dimension tables (DIM_DATE, DIM_COUNTRY, DIM_PRODUCT) from raw data.
- Removing duplicates, handling null values, and validating totals against official CAPMAS 2024 publications.

Key Data Tables

1. FACT_TRADE_FLOW — The Core Transaction Table

This is the central fact table of the entire Observatory. It contains Egypt's export and import records for every year from 2010 to 2024, broken down by product sector and trade type. With 3,870 rows, it is the most frequently queried table in the model.

Step	Transformation Applied
1. Import Source	Load CAPMAS annual trade data from Excel workbook into Power Query
2. Promote Headers	Convert first row into column headers using Table.PromoteHeaders()
3. Set Data Types	Assign "date" to Year column, "number" to Trade_Value_USD_M, "text" to Direction
4. Filter Direction	Separate Export and Import rows by filtering the Direction column
5. Remove Nulls	Table.SelectRows() to remove rows with missing trade values
6. Standardize Sectors	Text.Proper() to capitalize sector names consistently
7. Add Calculated Column	Add [is_petroleum] flag using if-then logic on sector names
8. Final Output	Clean, typed, structured fact table ready for data model relationships

2. DIM_DATE — The Calendar Dimension

The Date dimension is one of the most important tables in any Power BI model. It allows Power BI's time intelligence functions (like DATEADD, TOTALYTD, and SAMEPERIODLASTYEAR) to work correctly. Without a proper date table, year-over-year comparisons would fail.

For this project, the date table covers every day from January 1, 2010 to December 31, 2024 — 5,479 rows total. Each row includes the full date, year, quarter, month, fiscal year, a flag for crisis years (2016 EGP devaluation, 2020 COVID, 2022 commodity spike), and a Vision 2030 phase label.

3. DIM_COUNTRY — The Geographic Dimension

The country dimension provides a clean reference table for all 50+ countries that appear as Egypt's trade partners. Raw bilateral trade data from CAPMAS and UN Comtrade uses different country name conventions — "Kingdom of Saudi Arabia" vs. "Saudi Arabia" vs. "KSA." The DIM_COUNTRY table standardizes all names to a single canonical form and enriches each country with: continent, region, GDP tier (High/Middle/Low income), trade bloc membership (GAFTA, EU, COMESA, etc.), and World Bank region code.

4. FACT_EXPORT_UNIT_PRICE — Price Intelligence Table

This specialized fact table powers the entire Price Intelligence module (Page 5). It was sourced from ITC TradeMap and contains unit price data for Egypt's top 50 export products. For each product it holds: Egypt's average export price (USD/unit), the best-performing competitor country's price, the final importer-paid price, the price gap per unit, and the annual revenue loss calculation.

The annual revenue loss is calculated as: (Competitor Price − Egypt Price) × Egypt Export Volume. This is the number that produces the $3B+ figure displayed on the dashboard.

5. FACT_PEER_BENCHMARK — Competitive Intelligence Table

This table enables the benchmarking comparison on Page 6. It contains 14 standardized KPI values for Egypt, Turkey, Morocco, and Vietnam — all sourced from the World Bank and WTO 2024 datasets. Standardization was critical: some indicators use USD billions, others use index scores, and others use percentages. Each value was normalized to a common unit before loading.

09 Chapter 2: System Analysis & Design

1. Database Design & Data Modeling

What is a Data Model and Why Does It Matter?

A data model in Power BI is the structured arrangement of tables and the relationships between them. It determines how data flows when a user clicks a filter, applies a slicer, or drills into a visual. A poorly designed data model produces incorrect totals, slow dashboards, and confusing interactions. A well-designed model makes everything "just work."

The Observatory uses a Fact Constellation Schema — a design pattern where multiple fact tables share common dimension tables. This is the most appropriate architecture when an analytical project covers multiple distinct business processes (in our case: trade flows, price gaps, Vision KPIs, bilateral flows, and peer benchmarks), each requiring its own granularity.

The 17-Table Architecture

The data model consists of 7 fact tables and 6 dimension tables, connected by 9 defined relationships. Here is how to think about the distinction:

- Fact Tables contain numeric measurements — the "what happened" data. Examples: total exports by year, revenue lost by product, KPI achievement percentage.
- Dimension Tables contain descriptive attributes — the "context" data. Examples: country names and regions, product names and HS codes, date labels and fiscal years.
- Relationships connect fact tables to dimension tables — they are the "joints" that allow a filter on a dimension (e.g., "show me only 2024") to propagate correctly through all connected fact tables.

Fact Tables (7)

Table Name	Rows / Purpose
FACT_TRADE_FLOW	3,870 rows — Core export/import records 2010–2024 by sector and direction
FACT_BILATERAL_FLOWS	1,202 rows — Egypt's trade with each partner country for geographic intelligence (P3)
FACT_KPI_PROGRESS	140 rows — Annual Vision 2030 KPI achievement vs. target — backbone of P7
FACT_EXPORT_UNIT_PRICE	225 rows — Egypt price vs. competitor price — Price Intelligence engine (P5)
FACT_IMPORT_UNIT_PRICE	180 rows — Egypt import price vs. world average — import premium analysis
FACT_PRICE_GAP_SUMMARY	180 rows — Aggregated revenue loss by product — scenario engine input (P8)
FACT_PEER_BENCHMARK	60 rows — Egypt vs. Turkey / Morocco / Vietnam across 14 KPIs (P6)

Dimension Tables (6)

Table Name	Purpose & Key Columns
DIM_DATE	Full calendar 2010–2024. Columns: year, quarter, month, fiscal_year, is_crisis_year, vision_phase
DIM_COUNTRY	50 countries. Columns: country_name_en, continent, region, gdp_tier, trade_bloc
DIM_PRODUCT	50 products. Columns: product_name_en, hs_code, sector, complexity_index, is_petroleum, vision_priority
DIM_TRADE_TYPE	8 trade type classifications: Export/Import, General/Special, Free Zone
DIM_VISION2030_KPI	18 KPI definitions with baselines (2015), targets (2030), and priority levels
DIM_CHANNEL	14 logistics channel types: transport mode, port, and Incoterms classifications

2. UI/UX Design & Prototyping

Design Philosophy

The dashboard was designed following the principle that "the best analysis is the one that gets acted on." This means prioritizing clarity over complexity — every visual element must earn its place by either conveying information or guiding the user's attention.

Design Principle	How It Was Applied
5-Second Rule	The most important number on each page must be identifiable within 5 seconds of opening it.
Progressive Disclosure	KPI cards show the headline; custom tooltips reveal the detail on hover.
Consistent Navigation	Fixed left-panel navigation buttons appear on all 8 pages for frictionless movement.
Color Semantics	Navy = overview/structure, Gold = KPIs/highlights, Emerald = growth/positive, Red = deficit/alert.
Bilingual Labels	All page titles and KPI labels appear in both English and Arabic.

Dashboard Navigation Architecture

The navigation flows logically from strategic overview to operational detail: P1 Executive → P2 Trends → P3 Geography → P4 Products → P5 Prices → P6 Benchmarks → P7 Vision → P8 Scenarios. Each page has a fixed left-panel navigation bar and a filter panel at the bottom. Drill-through capabilities allow users to click on a country on the map (P3) and jump directly to that country's detailed bilateral flow analysis.

Color Palette & Typography

10 DAX Measures Library — 40+ Formulas Explained

What is DAX?

DAX (Data Analysis Expressions) is the formula language of Power BI. It works like Excel formulas, but is designed for relational data tables and complex analytical calculations. Every KPI card, trend line, and percentage you see in the dashboard is powered by a DAX measure.

All measures in this project live in a dedicated "_MEASURES" table. This keeps the data model clean — no measures are scattered across different tables. Add any measure via: Modeling → New Measure.

Section 1: Basic KPIs —

⚡ Total Exports USD M — Basic KPI

```
Total Exports USD M =
CALCULATE(
    SUM(FACT_TRADE_FLOW[trade_value_usd_million]),
    DIM_TRADE_TYPE[direction] = "Export"
)
```

What it calculates

Sums all trade values from FACT_TRADE_FLOW, but only for rows where the direction is "Export". The CALCULATE function applies the filter; SUM adds up the values.

Why it matters

This is the single most-used measure in the entire dashboard. It appears on the P1 KPI card, the waterfall chart, the trend chart, and 6 other pages. Without filtering to Export-only, the sum would incorrectly include imports.

⚡ Total Imports USD M — Basic KPI

```
Total Imports USD M =
CALCULATE(
    SUM(FACT_TRADE_FLOW[trade_value_usd_million]),
    DIM_TRADE_TYPE[direction] = "Import"
)
```

What it calculates

Identical logic to Total Exports but filtered to Direction = "Import". Returns the total USD value of all Egyptian imports for the selected time period.

Why it matters

Used on the P1 KPI card, the waterfall chart, and as an input to the Trade Balance and Trade Balance Ratio measures. Separating exports and imports into two clean measures is a best practice that prevents confusion and enables correct deficit calculation.

⚡ Trade Balance USD M — Basic KPI

```
Trade Balance USD M =
```

```
[Total Exports USD M] - [Total Imports USD M]
```

What it calculates

Subtracts total imports from total exports. When imports exceed exports (which is the case every year from 2010–2024), this measure returns a negative value — representing the trade deficit.

Why it matters

The Trade Balance is the headline metric of the entire Observatory. A persistent negative trade balance means Egypt is continuously consuming more foreign currency than it earns. The waterfall chart on P1 uses this measure to show how the deficit evolved from -$29B (2010) to -$50B (2024).

⚡ Non-Oil Export Share %

Structure KPI

```
Non-Oil Export Share % =
DIVIDE(
    CALCULATE(
        SUM(FACT_TRADE_FLOW[trade_value_usd_million]),
        DIM_TRADE_TYPE[direction] = "Export",
        DIM_PRODUCT[is_petroleum] = "No"
    ),
    [Total Exports USD M],
    0
) * 100
```

What it calculates

Calculates the percentage of total exports that come from non-petroleum products. The inner CALCULATE filters to Export AND non-petroleum rows; DIVIDE converts to a percentage; the "0" is the alternative result if Total Exports is zero (prevents division errors).

Why it matters

Vision 2030 sets a target of 75% non-oil export share. Egypt is currently at 55%. This measure appears as a KPI card on P1 and as a trend line on P2. Monitoring this KPI tells policymakers whether export diversification efforts are working or whether petroleum continues to dominate.

⚡ Count Active Partners

Diversification KPI

```
Count Active Partners =
CALCULATE(
    DISTINCTCOUNT(FACT_BILATERAL_FLOWS[partner_country_en]),
    FACT_BILATERAL_FLOWS[export_usd_m] > 0
)
```

What it calculates

Counts the number of distinct partner countries where Egypt has non-zero exports. DISTINCTCOUNT counts unique values; the filter ensures we only count countries with actual exports (not just trade agreements on paper).

Why it matters

Geographic diversification is one of Egypt's most urgent challenges. This measure appears on P3 (Geographic Intelligence) and provides context for the HHI concentration index. A higher count suggests better diversification, but the HHI is needed to understand whether exports are actually distributed across those partners.

Section 2: Growth Metrics —

⚡ YoY Export Growth %

Growth KPI

```
YoY Export Growth % =
VAR CurrentExports = [Total Exports USD M]
VAR PrevExports =
    CALCULATE(
        [Total Exports USD M],
        DATEADD(DIM_DATE[full_date], -1, YEAR)
    )
RETURN
    DIVIDE(
        CurrentExports - PrevExports,
        PrevExports,
        BLANK()
    ) * 100
```

What it calculates

Uses VAR (variables) to store two values: current year exports and prior year exports. DATEADD shifts the date context back by one year. DIVIDE calculates the growth percentage. BLANK() is returned if there is no prior year (e.g., for the first year in the dataset).

Why it matters

This is the foundational growth metric. It appears as a bar chart on P2 with color-coded bars (green for growth above CAGR, orange for below, red for contraction). It is critical for distinguishing the 2022 commodity-price spike (+34% YoY) from genuine structural export growth. Without this measure, the trend line alone would be misleading.

⚡ CAGR Exports 2010–2024

Long-term Growth

```
CAGR Exports 2010_2024 =
VAR ExportsStart =
    CALCULATE(
        [Total Exports USD M],
        DIM_DATE[year] = 2010
    )
VAR ExportsEnd =
    CALCULATE(
        [Total Exports USD M],
        DIM_DATE[year] = 2024
    )
VAR NumYears = 14
RETURN
    (POWER(DIVIDE(ExportsEnd, ExportsStart, BLANK()), 1/NumYears) - 1) * 100
```

What it calculates

CAGR = Compound Annual Growth Rate. The formula is: (End/Start)^(1/Years) - 1. POWER() performs the exponentiation. The result is a single percentage that represents the smoothed average annual growth rate over 14 years, removing the noise of year-to-year volatility.

Why it matters

Egypt's exports CAGR of 2.7% from 2010–2024 is compared directly against Vietnam's 14.2% and Morocco's 7.1% on the P6 benchmarking page. A 2.7% CAGR has been dramatically outpaced by peers, making this one of the most powerful "wake-up call" statistics in the dashboard.

⚡ Cumulative Export Growth Index

Index KPI

```
Cumulative Export Growth Index =
VAR BaseYear =
    CALCULATE(
        [Total Exports USD M],
        DIM_DATE[year] = 2010
    )
RETURN
    DIVIDE([Total Exports USD M], BaseYear, BLANK()) * 100
```

What it calculates

Sets 2010 = 100 and expresses every subsequent year as an index relative to 2010. If 2024 exports are 1.5x the 2010 level, the index shows 150. This allows direct comparison of growth trajectories across countries with different absolute scales.

Why it matters

Used on P6 to overlay Egypt's growth index against Turkey, Morocco, and Vietnam on the same chart — making it visually obvious that Vietnam's index has reached ~450 while Egypt's sits at ~145. Absolute dollar comparisons would be distorted by differences in starting size.

Section 3: Vision 2030 Tracking —2030

⚡ Vision Achievement %

Vision Tracker

```
Vision Achievement % =
AVERAGE(FACT_KPI_PROGRESS[achievement_pct])
```

What it calculates

Computes the simple arithmetic average of the achievement_pct column across all 18 Vision 2030 KPIs. This is the "overall score" for Egypt's Vision 2030 trade targets in any given year.

Why it matters

This measure powers the headline KPI card on P1 (currently showing 43%) and the progress bars on P7. It is the single most important summary statistic for Vision 2030 tracking — equivalent to a student's overall grade across all subjects. It appears in 4 different pages and is the first metric judges will ask about.

⚡ KPIs Critical Count

Vision Tracker

```
KPIs Critical Count =
CALCULATE(
    COUNTROWS(FACT_KPI_PROGRESS),
    FACT_KPI_PROGRESS[status] = "Critical",
    DIM_DATE[year] = 2024
)
```

What it calculates

Counts the number of KPI rows where the status field equals "Critical" in the year 2024. COUNTROWS counts records; the two filter conditions ensure we only count current-year critical KPIs.

Why it matters

This measure drives the "6 KPIs Critical" badge on P7. The number provides immediate urgency — it tells policymakers how many of the 18 Vision targets are at serious risk of being missed. Each critical KPI is then listed individually with its specific gap, giving a clear action list.

⚡ Years to 2030

Vision Tracker

```
Years to 2030 = 2030 - MAX(DIM_DATE[year])
```

What it calculates

Subtracts the most recent year in the dataset from the fixed target year of 2030. MAX(DIM_DATE[year]) dynamically returns the latest year available — currently 2024, giving a result of 6.

Why it matters

This deceptively simple measure creates enormous psychological impact. Displayed as a countdown in the top-right corner of P7, it constantly reminds users of the deadline. Combined with the Gap to Vision Target measure, it allows the dashboard to compute: "At the current growth rate, Egypt will reach $X by 2030 — falling $Y short of the $104B target."

Section 4: Price Intelligence —

⚡ Total Annual Revenue Lost

Revenue Intelligence

```
Total Annual Revenue Lost =
CALCULATE(
    SUM(FACT_PRICE_GAP_SUMMARY[total_gap_value_usd_m]),
    FACT_PRICE_GAP_SUMMARY[analysis_type] = "Export Underpricing"
)
```

What it calculates

Sums the pre-calculated annual revenue loss values from FACT_PRICE_GAP_SUMMARY, filtered to only "Export Underpricing" rows (as opposed to import premium rows). The revenue loss per product was pre-computed as: (Competitor Price − Egypt Price) × Egypt Export Volume.

Why it matters

The $3B+ figure this measure produces is arguably the single most impactful statistic in the entire Observatory. It demonstrates that Egypt could earn over $3 billion more per year — without producing or exporting a single additional unit — simply by aligning prices to market rates. This finding is what makes Recommendation R1 (Fix Export Pricing) the highest-priority action.

⚡ Avg Price Gap Export

Price Intelligence

```
Avg Price Gap Export =
AVERAGE(FACT_EXPORT_UNIT_PRICE[price_gap_usd_per_unit])
```

What it calculates

Averages the price gap column across all export products. The price_gap_usd_per_unit column contains: Competitor_Price − Egypt_Price for each product row. A positive gap means Egypt is selling below the competitor benchmark.

Why it matters

Used on P5 as a KPI card and as the sorting criterion for the revenue loss ranking bar chart. This measure answers: "On average, how much less per unit is Egypt getting vs. competitors?" Combined with volume data, it produces the revenue loss projection.

⚡ Circular Trade Count

Trade Intelligence

```
Circular Trade Count =
CALCULATE(
    COUNTROWS(FACT_IMPORT_UNIT_PRICE),
    FACT_IMPORT_UNIT_PRICE[circular_trade_flag] = "YES - Circular Trade!"
)
```

What it calculates

Counts the number of products where Egypt both exports the raw form cheaply AND imports the processed form expensively. This is called "circular trade" — Egypt is essentially selling cheap and buying back expensive, with the value-add captured abroad.

Why it matters

The classic example: Egypt exports raw cotton at ~$800/ton, then imports cotton fabrics from China at ~$2,841/ton. This measure quantifies how widespread this inefficiency is. Finding 8+ circular trade products is strong evidence for the import substitution recommendations (R7 in the strategy section).

Section 5: Benchmarking —

⚡ Egypt vs. Vietnam Gap %

Benchmark KPI

```
Egypt vs. Vietnam Gap % =
VAR EgyptVal =
    CALCULATE(
        SUM(FACT_KPI_BENCHMARKS[egypt_2024]),
        FACT_KPI_BENCHMARKS[indicator_name_en] = "Total Exports (USD Billion)"
    )
VAR VietnamVal =
    CALCULATE(
        SUM(FACT_KPI_BENCHMARKS[vietnam_2024]),
        FACT_KPI_BENCHMARKS[indicator_name_en] = "Total Exports (USD Billion)"
    )
RETURN
    DIVIDE(VietnamVal - EgyptVal, EgyptVal, 0) * 100
```

What it calculates

Retrieves Egypt's and Vietnam's 2024 export values from the benchmark table using CALCULATE with indicator name filters. DIVIDE computes the percentage by which Vietnam exceeds Egypt. The "wake-up call" metric: Vietnam exports roughly 11× more than Egypt.

Why it matters

Vietnam had a lower GDP per capita than Egypt in 2000 and lacked Egypt's geographic advantages (no Suez Canal, no MENA market access). Yet today Vietnam exports $400B+ vs. Egypt's $32.6B. This gap — quantified by this measure — is the most powerful argument for structural reform in the entire dashboard.

⚡ Geographic HHI

Diversification Index

```
Geographic HHI =
VAR TotalExp = SUMX(
    SUMMARIZE(FACT_BILATERAL_FLOWS,
              FACT_BILATERAL_FLOWS[partner_country_en],
              "PartnerExp", SUM(FACT_BILATERAL_FLOWS[export_usd_m])),
    [PartnerExp]
)
RETURN
SUMX(
    SUMMARIZE(FACT_BILATERAL_FLOWS,
              FACT_BILATERAL_FLOWS[partner_country_en],
              "PartnerExp", SUM(FACT_BILATERAL_FLOWS[export_usd_m])),
    POWER(DIVIDE([PartnerExp], TotalExp, 0), 2)
)
```

What it calculates

HHI = Herfindahl-Hirschman Index. SUMMARIZE groups exports by partner country. SUMX iterates over each partner, squares its share of total exports (POWER(..., 2)), and sums the squared shares. The result ranges from 0 (perfect diversification) to 1 (all exports to one country).

Why it matters

HHI is the gold standard for measuring market concentration in economics. The U.S. Department of Justice classifies markets above 0.25 as "highly concentrated." Egypt's geographic HHI is ~0.22 — approaching that threshold. This measure provides the statistical backing for the geographic diversification recommendations, making the argument rigorous rather than subjective.

Section 6: Geographic Intelligence —

⚡ Top 5 Products Share %

Concentration KPI

```
Top5 Products Share % =
VAR Top5 =
    TOPN(5,
         SUMMARIZE(FACT_TRADE_FLOW,
                   DIM_PRODUCT[product_name_en],
                   "ProdExp", [Total Exports USD M]),
         [ProdExp], DESC)
VAR Top5Value = SUMX(Top5, [ProdExp])
RETURN
    DIVIDE(Top5Value, [Total Exports USD M], 0) * 100
```

What it calculates

SUMMARIZE groups total exports by product. TOPN selects the top 5 by export value. SUMX sums those top 5 values. DIVIDE converts to a percentage of total exports. This is a product-level concentration index equivalent to the geographic HHI.

Why it matters

If the top 5 products account for 60–70% of total exports, the economy is dangerously dependent on a handful of products. This measure supports the export diversification argument on P4 and benchmarks against peers (Vietnam's top 5 products represent a much smaller share of its total exports, demonstrating the value of an industrialized, diversified export base).

⚡ Export Gap to 2030 Target

Vision Gap KPI

```
Export Gap to 2030 Target =
VAR Target2030 = 104
VAR Current =
    CALCULATE(
        [Total Exports USD M] / 1000,
        DIM_DATE[year] = 2024
    )
RETURN Target2030 - Current
```

What it calculates

Sets the Vision 2030 target at $104 billion (converted from USD billions). Calculates current 2024 exports in USD billions (dividing the USD millions figure by 1,000). Subtracts to find the remaining gap. Currently returns ~71.4 (i.e., Egypt needs $71.4B more in exports to reach the target).

Why it matters

This measure, displayed prominently on P7 and P8, creates the "burning platform" for the entire strategy section. A $71.4B gap with 6 years remaining requires extraordinary structural change — not incremental improvement. It is the mathematical foundation for all 12 strategic recommendations.

Section 7: Advanced Analytics —

⚡ WhatIf Revenue Gain — Scenario KPI

```
WhatIf Revenue Gain =
CALCULATE(
    SUM(FACT_PRICE_GAP_SUMMARY[total_gap_value_usd_m]),
    FACT_PRICE_GAP_SUMMARY[gap_severity] = "Critical"
)
```

What it calculates

Sums the revenue gap for products classified as "Critical" severity — products where the pricing gap is largest and the volume is highest. This is the baseline calculation for the What-If engine. The interactive slicer on P8 multiplies this by a user-selected improvement percentage.

Why it matters

The What-If engine on P8 is the most sophisticated feature in the dashboard. By combining this measure with a Power BI numeric parameter slicer, the dashboard becomes interactive: users can set "assume Egypt captures 40% of the pricing gap" and see the revenue impact update in real time. This transforms the dashboard from a reporting tool into a strategic planning tool.

⚡ Exports per Capita — Per-Capita KPI

```
Exports per Capita =
DIVIDE(
    [Total Exports USD M] * 1000000,
    CALCULATE(MAX(FACT_PEER_BENCHMARK[population_millions]),
              FACT_PEER_BENCHMARK[country_name_en] = "Egypt") * 1000000,
    0
)
```

What it calculates

Converts total exports from USD millions to USD by multiplying by 1,000,000. Retrieves Egypt's population in millions from the benchmark table and similarly converts to individuals. DIVIDE produces exports per person in USD. This allows fair comparison across countries of different sizes.

Why it matters

Egypt's exports per capita (~$310) are dramatically below Turkey (~$2,700) and Vietnam (~$4,100). This normalization is critical: Turkey's higher absolute exports partly reflect a larger economy, but per-capita comparison reveals structural differences in export intensity. This measure appears on P6 and is one of the most impactful visuals for benchmark audiences.

11 Non-Functional Requirements

Non-functional requirements define HOW the system works — its quality attributes — as opposed to functional requirements which define WHAT it does. For a Power BI dashboard, these requirements govern performance, accuracy, usability, and maintainability.

Requirement	Specification
Performance	All 8 pages load and render within 3 seconds on a standard business laptop. DAX measures execute in under 500ms under typical slicer interactions. Optimized using VAR patterns and minimized bidirectional relationships.
Scalability	Data model accommodates additional years (post-2024) and new fact tables without schema redesign. Dimension tables support expansion to 100+ countries and 150+ products.
Usability	All navigation, drill-through, bookmark toggles, and What-If slicers function without technical guidance. Bilingual labels (English/Arabic) on all pages. Tooltip descriptions explain each visual for first-time users.
Accuracy	All KPI figures validated against CAPMAS 2024 published totals. No estimated or interpolated values displayed without a visible label. DAX measures independently verified by two team members.
Accessibility	Minimum 4.5:1 contrast ratio throughout. Critical information conveyed through both color and label/shape (color-blind accessible design).
Maintainability	All DAX measures centralized in _MEASURES table. All transformations documented in Power Query step descriptions. Blueprint Excel file updated to reflect any schema changes.
Compatibility	Dashboard renders correctly on Power BI Desktop (Jan 2025 release), Power BI Service (web), and projection screens at 1920×1080 (16:9 canvas ratio).
Security	All data files and outputs classified as Confidential — restricted to DEPI evaluation use only.

12 Risks & Assumptions

Risk Register

Risk / Assumption	Category	Impact	Mitigation
Data Gaps (Pre-2015)	**High**	High	Use available data from 2010; clearly label interpolated values and flag known gaps in tooltips.
CAPMAS Revision Lag	**Data Quality**	Medium	Use most recently published official CAPMAS 2024 data; note publication date on all KPI cards.
Exchange Rate Volatility	**Methodology**	Medium	All values reported in USD millions at published exchange rates; methodology documented in data model sheet.
Power BI Service Performance	**Technical**	Low	Optimized on Desktop; minimize bidirectional relationships and use VAR patterns in all DAX measures.
Benchmarking Data Currency	**Data Quality**	Medium	Peer data sourced from WTO and World Bank 2024 publications; flagged when 2023 data is used as proxy.
Vision 2030 Target Revision	**Policy**	Low	KPI targets locked at official 2023 Vision 2030 document values; revisions require model update.
Scope Creep	**Project Management**	Medium	Out-of-scope items formally documented; change requests require supervisor approval before implementation.

Project Assumptions

- All data from CAPMAS, UN Comtrade, WTO, World Bank, and ITC TradeMap is accepted as accurate as of the publication dates cited.
- The Vision 2030 export target of $104 billion refers to total goods exports; services exports are tracked separately and not included in the primary KPI.
- Competitor pricing benchmarks (P5 Price Intelligence) reflect the most recent ITC TradeMap unit price data at HS-code level.
- The HHI calculations use bilateral export flows only; re-export and transit trade are excluded.
- Power BI Desktop January 2025 release is the target build environment.
- The DEPI evaluation panel will have access to a fully functional Power BI Desktop environment for live demonstration.

13 Project Timeline & Milestones

The project follows a 10-phase structured timeline. Each phase has a clear input, activity, and deliverable — ensuring accountability and enabling the team to course-correct if a phase runs long.

Phase	Activity	Duration	Deliverable / Milestone
01	**Problem Definition & Data Discovery**	Week 1–2	Approved problem statement · Data audit report · Source validation
02	**Data Collection & Integration**	Week 2–3	17 structured tables loaded and validated in Power Query
03	**Data Modeling & Schema Design**	Week 3–4	Fact Constellation Schema approved · All 9 relationships established
04	**DAX Measure Authoring**	Week 4–5	40+ DAX measures validated against published CAPMAS 2024 totals
05	**Dashboard Design & Wireframing**	Week 5–6	Layout system finalized across all 8 pages
06	**Dashboard Build — Pages P1–P4**	Week 6–7	4 pages with full interactivity, custom tooltips, and bilingual labels
07	**Dashboard Build — Pages P5–P8**	Week 7–8	Price Intelligence, Benchmarking, Vision Tracker, and What-If engine
08	**QA, Validation & Performance Testing**	Week 8–9	All figures validated · Load time confirmed < 3 seconds per page
09	**Script Development & Rehearsal**	Week 9	Full bilingual presentation script · 5 complete rehearsal runs
10	**Final Submission & Live Presentation**	Week 10	Dashboard submitted · Live presentation delivered to DEPI panel

14 Chapter 3: Analysis & Insights

Key Findings from the Data

Finding 1: The Trade Deficit Is Structural, Not Cyclical

Egypt's trade deficit expanded from -$29B (2010) to -$50B (2024), representing a 72% widening in 14 years. The 2022 spike to a temporary improvement was driven entirely by LNG and petroleum price surges — when commodity prices normalized in 2023, the deficit returned. This confirms the deficit is structural (driven by the composition of exports) rather than cyclical (driven by temporary price movements). No short-term monetary intervention can fix a structural trade deficit; only export diversification and value-addition can.

Finding 2: Petroleum Dependency Has Not Improved

In 2010, petroleum accounted for 63% of exports. In 2024, it accounts for 62% — essentially unchanged over 14 years despite stated policy goals of diversification. Meanwhile, non-oil exports grew at only 1.9% CAGR — barely ahead of inflation. The manufacturing sector, which typically drives export transformation (as seen in Vietnam, Turkey, and Morocco), accounts for only 12% of Egypt's non-oil exports.

Finding 3: $3.2B in Annual Pricing Losses Are Recoverable

The price gap analysis reveals that Egypt is not simply selling the wrong products — it is selling the RIGHT products at the WRONG price. Citrus, cotton, marble, chemicals, and fertilizers all represent categories where Egypt has a genuine Revealed Comparative Advantage (RCA > 1) but systematically sells 30–70% below market rates. The top 10 most affected products account for over $2.8B of the $3.2B total annual loss.

Finding 4: Circular Trade Is Costing Egypt Twice

Analysis of FACT_IMPORT_UNIT_PRICE identified 8+ product categories where Egypt exports the raw material cheaply and imports the processed equivalent expensively. The cotton example is the most striking: Egypt exports raw long-staple cotton at ~$800/ton, then pays ~$2,841/ton to import cotton fabrics. Egypt is effectively subsidizing foreign textile industries while missing the manufacturing value-add that would create domestic jobs and higher export revenues.

Finding 5: Vietnam's Trajectory Is Egypt's Roadmap

The benchmarking analysis reveals that Vietnam's export transformation — from ~$15B in 2000 to $400B+ today — was not accidental. It was driven by three deliberate policy choices: attracting manufacturing FDI with generous EPZ incentives; rapidly expanding export market diversification (Vietnam today exports to 200+ countries); and investing in logistics infrastructure (LPI improved from 2.1 to 3.3 in 15 years). All three are directly addressable by Egypt with lower political resistance than structural reforms typically face.

Executive Summary of Findings

★ Trade deficit widened from -$29B to -$50B — structural, not cyclical, requiring export transformation.
★ Petroleum share unchanged at ~62% over 14 years despite diversification policy goals.
★ $3.2B in annual export revenue recoverable through pricing correction alone — no volume increase needed.
★ 8+ circular trade products identified — Egypt selling cheap, buying back expensive.
★ Vietnam's 14-year export journey provides a concrete, evidence-based reform roadmap for Egypt.
★ Vision 2030 achievement at 43% overall; 6 of 18 KPIs classified as Critical.

15 Strategic Recommendations — 12 Data-Driven Actions

All 12 recommendations are evidence-based (derived directly from the Observatory's data), impact-quantified (projected revenue gain or cost saving), and time-bound (short/medium/long-term classification).

R1 Fix Export Pricing — CRITICAL — 2–3 Yrs

Evidence: Egypt sells citrus at $333/ton; Spain achieves $426/ton for the same variety. Across all products, this pricing gap costs Egypt $3.2B+ annually.

Projected Impact: +$3.2B annually — pure margin improvement, no volume change required.

R2 National Export Branding Program — CRITICAL — 3–5 Yrs

Evidence: Turkey's TURQUALITY program boosted branded exports by 38% in 4 years. Egypt's Giza Cotton sells unbranded at $3,730/ton vs. $6,096 for branded equivalents.

Projected Impact: +$4.8B export value by 2030 for top 5 branded Egyptian products.

R3 Structural Export Diversification — CRITICAL — 5–7 Yrs

Evidence: Every 10% structural shift from petroleum to non-petroleum adds $3B+ in stable, non-volatile revenue.

Projected Impact: Reduces vulnerability to oil price cycles. Vision 2030 target: 75% non-oil.

R4 Fast-Track EU Full Free Trade Agreement — CRITICAL — 3–5 Yrs

Evidence: Turkey's EU Customs Union (1996) triggered a 42% export jump within 5 years. Egypt's current Association Agreement falls far short.

Projected Impact: +$8.5B potential exports to EU within 5 years based on Turkey precedent.

R5 Asian Market Export Strategy — HIGH — 3–5 Yrs

Evidence: Asia = 60% of global GDP growth. Egypt sends only 18% of exports there.

Projected Impact: Doubling Asia's share from 18% to 36% adds +$5.8B annual export revenue.

R6 Export Processing Zones (Textiles & Pharma) — HIGH — 3–5 Yrs

Evidence: Vietnam's EPZs drove $150B+ in manufacturing exports. Morocco's textile EPZ generated $5B in 8 years.

Projected Impact: +$3.6B garment and pharmaceutical exports by 2030.

R7 Import Substitution — 3 Key Sectors — HIGH — 2–4 Yrs

Evidence: Egypt exports raw cotton and imports processed cotton fabrics at $2,841/ton. 8+ circular trade products identified.

Projected Impact: +$1.8B annual savings by substituting the top 3 circular trade products.

R8	Improve Logistics Performance Index to 3.5	HIGH	3–4 Yrs

Evidence: Egypt LPI = 2.82 vs. Turkey 3.62, Vietnam 3.27. Each 0.1 LPI improvement = +3% export growth (World Bank).

Projected Impact: +4–6% annual export boost = $1.5–2B in additional exports per year.

R9	Scale Food Export Value Chain	MEDIUM	4–6 Yrs

Evidence: Egypt food export target: $12B by 2030 (baseline: $3.2B). Citrus, tomatoes, herbs: underpriced and under-certified.

Projected Impact: +$8.8B in food export revenue with cold chain and certification investment.

R10	Expand ICT & Digital Exports	MEDIUM	5–7 Yrs

Evidence: ICT export target: $5B by 2030 (baseline: $0.3B). Large tech talent pool, no export-oriented digital services framework.

Projected Impact: +$4.7B in ICT exports — highest complexity, highest margin sector available.

R11	Enhance Suez Canal Revenue to $13.5B	MEDIUM	3–7 Yrs

Evidence: Post-Red Sea disruption recovery target: $13.5B by 2030 (current: ~$7.5B).

Projected Impact: +$6B Suez revenue through logistics hub and ship repair services expansion.

R12	Attract $20B Manufacturing FDI by 2030	CRITICAL	3–7 Yrs

Evidence: Vietnam: each $1B manufacturing FDI generates ~$3B in exports within 5 years. Egypt's current FDI: $10.1B.

Projected Impact: Each $1B manufacturing FDI → ~$3B in exports. Target: $20B FDI by 2030.

16 KPIs & Success Metrics

Analytical KPIs (Dashboard Outputs)

KPI Measure	Definition & Target
Total Exports USD M	$32.6B in 2024. Vision 2030 target: $104B. Required CAGR from 2024: 18%.
Trade Balance USD M	Exports minus Imports. Current: -$50B. Progressive target: reduce to -$20B by 2030.
Non-Oil Export Share %	Currently 55%. Vision 2030 target: 75%.
Vision KPI Achievement %	Average across 18 KPIs. Currently 43%. Target: 100% by 2030.
Total Annual Revenue Lost	$3.2B currently. Target: <$500M after pricing corrections.
Egypt vs. Vietnam Gap %	Vietnam exports ~11× more. Target: reduce gap to <5× by 2030.
Geographic HHI	~0.22 currently (approaching concentrated). Target: <0.15.
YoY Export Growth %	Required from 2024 base: 18% per annum to reach Vision 2030 target.
Manufacturing Export Share %	~12% currently. Vision 2030 target: 60%.

Project Quality Metrics (Delivery)

Quality Metric	Target
Dashboard Page Count	8 fully functional pages (P1–P8) delivered on time
DAX Measure Count	Minimum 40 validated measures across 7 analytical categories
Data Validation Rate	100% of KPIs validated against official CAPMAS 2024 published figures
Dashboard Load Time	< 3 seconds for all 8 pages on standard business hardware
Custom Tooltip Coverage	7 custom tooltip pages covering all major visual types
Bilingual Coverage	100% of titles and KPI labels in both English and Arabic
Interactive Feature Count	Minimum 8 features (drill-through, bookmarks, parameters, cross-filter)
Recommendation Count	12 data-driven recommendations with quantified impact and timeline

17 Project Team

The Egyptian Export Observatory was developed by a five-member team convened under the Digital Egypt Pioneers Initiative (DEPI). Our team brings together expertise spanning economics, business intelligence, data engineering, trade policy, and financial analysis. Below is the full team roster with LinkedIn profiles for professional connection.

#	Full Name	LinkedIn Profile
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Digital Egypt Pioneers Initiative (DEPI) | Data Analysis Track | Egypt Trade Intelligence Dashboard

"The data is clear. The path is mapped. The question is: who takes action?"

— Egyptian Export Observatory, DEPI Graduation Project 2025